

An Introduction to the Mathematical Theory and Core Concepts of the Theory of Entropicity

Orientation and scope

This paper presents Obidi's Theory of Entropicity as a **programmatic research framework** rather than as a completed, universally accepted theory. Its aim is to show, as clearly as possible, how the mathematical objects already used in information geometry, differential geometry, quantum information, kinetic theory, and general relativity can be assembled into a coherent entropic-first picture of physics. The guiding idea is the one laid out in your attached *Letter III*: entropy is not treated as a late thermodynamic bookkeeping device, but as the primitive field whose organized gradients, localization, and coarse-graining give rise to geometry, matter, and dynamical law. The goal of this monograph-grade introduction is therefore twofold: first, to separate what is already standard mathematics from what is specific to ToE; second, to show why the ToE route from information geometry to spacetime geometry, and from entropic distributions to the Einstein right-hand side, is mathematically intelligible. [filecite turn0file0](#)

The most important conceptual discipline is this: **not every mathematically definable entropy field is automatically physical spacetime, and not every probability distribution is automatically mass or pressure**. What ToE needs to show is a chain of maps. First, a space of distinguishable states must acquire a metric. Second, that metric must be converted from a positive-definite information metric into a Lorentzian causal metric. Third, localized entropic distributions must produce rank-two source tensors whose low-order moments reproduce density, momentum flux, pressure, anisotropic stress, and radiation. Fourth, the geometric side and the source side must satisfy matched conservation laws, just as the Einstein tensor and stress-energy tensor do in standard general relativity. That is exactly the program announced in *Letter III*, and it is the program clarified here. [filecite turn0file0](#)

It is also important to place ToE in the broader research landscape. Modern theoretical physics already contains several influential bridges between entropy, information, and gravity: Jacobson's derivation of the Einstein equation as an equation of state; Verlinde's entropic-force program; Caticha's entropic dynamics on statistical manifolds; Matsueda's use of Fisher geometry to obtain Einstein-like structures; and Bianconi's recent "gravity from entropy" action built from quantum relative entropy between geometric objects. ToE belongs in that family of ideas, but it makes a sharper claim than most of them: it seeks a unified route from **entropic information geometry** to **physical Lorentzian spacetime** and then to **both sides** of the Einstein field equations. ¹

The revolutionary inversion

In standard textbook physics, matter and radiation are taken as fundamental enough that one first writes down particles, fields, energies, momenta, and interactions, and only later computes entropy from the multiplicity or uncertainty of those states. ToE inverts that order. It says: begin with entropy, understood as structured distinguishability over a manifold of possible states; then let geometry, matter content, and dynamics emerge from how that entropy is distributed, transported, and stabilized. This is what makes the

theory feel revolutionary: it changes the explanatory direction. The inversion is bold, but it is not logically absurd, because modern physics already recognizes that information has physical content, that entropy can be localized in non-equilibrium settings, and that gravity itself may admit thermodynamic or informational formulations. ²

A field, in mathematics, simply means that to every point of some base space we assign an object. If we assign one real number to every point, we have a **scalar field**. If we assign an arrow-like object to every point, we have a **vector field**. If we assign a linear measuring device to every point, we have a **covector field** or **one-form**. Therefore, from a purely mathematical point of view, “entropy is a field” means only that there is a function

$$S : M \rightarrow \mathbb{R}$$

or more generally a locally defined entropy density/current over a manifold M . This is not exotic language. In non-equilibrium thermodynamics and relativistic thermodynamics one already uses local entropy densities and entropy currents. So ToE does not invent the idea that entropy can be locally assigned; its novelty is to elevate that local structure from a secondary thermodynamic variable to the ontological starting point. ³

To understand why this matters physically, it helps to distinguish three levels. At the first level, entropy is a scalar $S(x)$: one number at each point. At the second, its gradient

$$dS = \partial_\mu S dx^\mu$$

is a covector field telling us the direction in which entropy increases fastest. At the third, once a metric exists, this gradient can be turned into a vector

$$(\nabla S)^\mu = g^{\mu\nu} \partial_\nu S,$$

which can then define preferred flow lines. In ordinary language, the scalar S tells you “how much”, the gradient dS tells you “which way it changes most strongly”, and the metric tells you how to convert this information into distances, norms, and causal structure. ToE’s proposal is that the timelike direction of spacetime is ultimately selected by entropy flow. ⁴

This is why ToE insists that mass, pressure, radiation, and stress are not alien to information; they are what highly organized information looks like after localization and coarse-graining. A gas in equilibrium is not “made of pressure”; it is made of microscopic states whose collective statistics possess a second moment that we macroscopically call pressure. Radiation is not “made of luminosity”; it is a distribution of excitations whose momentum flux becomes a rank-two tensor. Mass density is not a primitive number that floats free of statistics; in kinetic theory it arises from the appropriate moments of a distribution on a phase-space bundle. So the ToE intuition that density, pressure, momentum flux, and stress are concentrated or stabilized information is not hand-waving. It becomes mathematically meaningful exactly when one introduces the right bundles, measures, and moments. ⁴

The mathematical language of distinguishability

The basic arena of information geometry is a **statistical manifold**. A point on that manifold is not an ordinary location in space, but a probability distribution $p(x|\theta)$ labeled by parameters $\theta = (\theta^1, \dots, \theta^n)$.

Two nearby points correspond to two nearby probability laws. The central question then becomes: how distinguishable are they? The answer is encoded by divergence functions and their second-order expansions. Shannon's entropy,

$$H(p) = - \sum_i p_i \log p_i,$$

measures uncertainty in a single distribution. The Kullback–Leibler divergence,

$$D_{\text{KL}}(p||q) = \sum_i p_i \log \frac{p_i}{q_i},$$

measures the information loss incurred when q is used in place of p . It is not symmetric, so it is not a metric, but its Hessian generates one. ⁵

That metric is the **Fisher–Rao metric**:

$$g_{ij}^{\text{F}}(\theta) = \mathbb{E}_{\theta}[\partial_i \log p(X|\theta) \partial_j \log p(X|\theta)] = -\mathbb{E}_{\theta}[\partial_i \partial_j \log p(X|\theta)].$$

Its geometric meaning is simple: if this metric is large in some direction of parameter space, then a small change in parameters causes a big change in the distribution, so states in that direction are easy to distinguish. If it is small, the distributions are hard to distinguish. Infinitesimally, the KL divergence expands as

$$D_{\text{KL}}(p_{\theta+d\theta}||p_{\theta}) = \frac{1}{2} g_{ij}^{\text{F}}(\theta) d\theta^i d\theta^j + O(d\theta^3).$$

So the Fisher–Rao metric is literally the second-order distinguishability geometry of a family of probability models. Čencov's theorem then says, in essence, that among Riemannian metrics respecting sufficient-statistics monotonicity, the Fisher metric is unique up to scale. ⁶

A statistical manifold also carries more structure than just one metric. Amari's α -connections describe a whole family of affine geometries compatible with the same statistical data. In a convenient lowered-index form one writes

$$\Gamma_{ijk}^{(\alpha)} = \Gamma_{ijk}^{(0)} - \frac{\alpha}{2} T_{ijk},$$

where $\Gamma^{(0)}$ is the Levi–Civita connection of g^{F} and

$$T_{ijk} = \mathbb{E}_{\theta}[\partial_i \log p \partial_j \log p \partial_k \log p]$$

is the cubic statistical tensor. The case $\alpha = 0$ recovers the ordinary Levi–Civita connection; $\alpha = \pm 1$ yields the dual exponential and mixture connections that are central in information geometry. In plain language, the metric measures local distinguishability, while the α -connections tell you how neighboring distributions are “stitched together” globally when you move across the manifold. In ToE, this family becomes physically suggestive because non-equilibrium, irreversibility, and asymmetric entropy flow can be interpreted as geometrically encoded in $\alpha \neq 0$. The specific constitutive law

$$\alpha = 2(1 - q)$$

appearing in *Letter III* should be understood as a **ToE postulate**, not as a standard theorem of information geometry: it ties non-extensive entropic behavior measured by q to geometric duality measured by α .

filecite turn0file0

The quantum analogue is the **Fubini–Study metric** on projective Hilbert space, the space of rays $[\psi]$ rather than vectors $|\psi\rangle$. For normalized pure states it can be written, up to normalization convention, as

$$ds_{\text{FS}}^2 = 4\left(\langle d\psi|d\psi\rangle - |\langle\psi|d\psi\rangle|^2\right).$$

Its meaning is again distinguishability: it measures how much a quantum state really changes once the physically irrelevant overall phase has been removed. In quantum metrology and quantum information, this is tightly linked to quantum Fisher information. Thus the classical Fisher–Rao metric and the quantum Fubini–Study metric are two faces of the same broad idea: geometry as infinitesimal distinguishability. ToE uses both because its deepest ambition is to make classical spacetime and quantum state structure emerge from one entropic substrate. ⁷

The quantum divergence that plays the role of KL divergence is the **Umegaki relative entropy**,

$$D_{\text{AU}}(\rho\|\sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)].$$

It measures distinguishability between quantum states ρ and σ , is non-negative, and is monotone under noisy quantum channels. This quantity matters for ToE because Bianconi’s recent entropic gravity program explicitly builds an action from a quantum relative entropy between geometric objects, and your *Letter III* uses the KL/Araki–Umegaki family as part of its generalized divergence language. That is a mathematically serious move: it places ToE in contact not only with classical statistics but with operator-algebraic quantum information. ⁸

At this point the elementary differential-geometric language must be completely clear. If M is a manifold and $x \in M$, then the **tangent space** $T_x M$ is the vector space of possible infinitesimal directions one can move through x . The **cotangent space** $T_x^* M$ is the space of linear functionals on those directions; it is where gradients naturally live. The **tangent bundle** $TM = \bigsqcup_x T_x M$ collects all tangent spaces together, point by point. The **cotangent bundle** $T^*M = \bigsqcup_x T_x^* M$ does the same for covectors. A metric is what lets one convert vectors into covectors and back. A gradient is therefore born as a covector, while a flow is naturally a vector. This is exactly why the entropy scalar S , the entropy gradient dS , and the entropy flow $(\nabla S)^\mu$ are three different but connected mathematical objects. ⁹

From information geometry to Lorentzian spacetime

Here we reach the central obstruction that *Letter III* correctly identifies. The Fisher–Rao metric is positive-definite on ordinary statistical manifolds, and the Fubini–Study metric is likewise Riemannian on projective Hilbert space. But physical spacetime in relativity is not Riemannian; it is **Lorentzian**. It has one timelike direction and three spacelike directions. This means that no mere change of coordinates on a positive-definite Fisher manifold can magically produce the causal structure of relativity. A genuine information-to-spacetime theory must therefore introduce **additional structure** that can change signature rather than merely re-label coordinates. That logical point is one of the strongest motivations for the ToE-specific “Obidi transformation.” filecite turn0file0

A mathematically clean version of that proposal is this. Let G_{AB} be a positive-definite information metric, for example Fisher–Rao, on an information manifold $\mathcal{I}$. Let S be the entropy scalar field on $\mathcal{I}$, and assume $dS \neq 0$. Define the normalized entropy-gradient one-form

$$u_A = \frac{\partial_A S}{\sqrt{G^{CD} \partial_C S \partial_D S}}.$$

Then define the **Obidi metric**

$$\tilde{G}_{AB} = \Omega^2 (G_{AB} - 2u_A u_B),$$

where Ω is an optional conformal factor. Because u_A has unit norm with respect to G , the update subtracts twice the projector along one distinguished entropic direction. In an orthonormal frame where $u_A = (1, 0, \dots, 0)$, one gets

$$G_{AB} = \text{diag}(1, 1, \dots, 1) \implies \tilde{G}_{AB} = \text{diag}(-1, 1, \dots, 1).$$

So one positive eigenvalue flips sign, and a Lorentzian signature appears. By the matrix determinant lemma,

$$\det \tilde{G} = \Omega^{2n} \det(G) (1 - 2) = -\Omega^{2n} \det(G),$$

which is exactly the expected sign flip for a single timelike direction. In short: the Obidi transformation is a **rank-one sign-flip along the entropy gradient**. [fileciteopen0file0](#)

This proposal is mathematically reminiscent of known gradient-dependent metric deformations in gravitational theory, especially disformal transformations of the Bekenstein type, where a metric is modified by terms built from a scalar field gradient. The ToE move is not identical to those constructions, but it belongs to that family of ideas: one starts from a background metric and then singles out a preferred scalar-gradient direction to alter causal structure. That analogy is useful because it shows the Obidi transformation is not an arbitrary fantasy; it is a sharper entropic variant of a recognized geometric strategy. ¹⁰

A simple worked example makes the point vivid. Consider the one-dimensional Gaussian family $p(x|\mu, \sigma)$. Its Fisher metric is

$$G = \frac{1}{\sigma^2} d\mu^2 + \frac{2}{\sigma^2} d\sigma^2.$$

The Shannon entropy of the Gaussian is

$$S(\mu, \sigma) = \frac{1}{2} \log(2\pi e \sigma^2),$$

so dS points purely in the σ -direction. Applying the Obidi construction flips the sign of that entropic direction and yields

$$\tilde{G} = \frac{1}{\sigma^2} d\mu^2 - \frac{2}{\sigma^2} d\sigma^2,$$

which is already Lorentzian in two dimensions. Physically, that means one distinguishability direction has been reinterpreted as timelike because it is aligned with entropy flow. This is exactly the kind of local demonstration that makes the ToE mechanism intelligible. [filecite turn0file0](#)

Once a Lorentzian metric exists, the rest of the left-hand side of general relativity follows by standard differential geometry. One computes the Levi-Civita connection $\tilde{\Gamma}^A_{BC}$, the Riemann tensor $\tilde{R}^A_{BCD}$, the Ricci tensor $\tilde{R}_{AB}$, the scalar curvature $\tilde{R}$, and the Einstein tensor

$$\tilde{G}_{AB} = \tilde{R}_{AB} - \frac{1}{2}\tilde{G}_{AB}\tilde{R}.$$

The contracted Bianchi identity then guarantees

$$\tilde{\nabla}^A \tilde{G}_{AB} = 0.$$

This is not a special extra assumption; it is automatic once one really has an Einstein tensor built from a metric connection. Therefore, if ToE successfully produces a genuine Lorentzian metric from entropic data, then the divergence-free structure of the geometric left-hand side comes for free. That is one of the strongest mathematical advantages of the Obidi strategy. ¹¹

To connect information space to physical spacetime, one then uses an emergence map $x \mapsto \theta(x)$, so that the spacetime metric is the pullback

$$g_{\mu\nu}(x) = \partial_\mu \theta^A \partial_\nu \theta^B \tilde{G}_{AB}(\theta(x)).$$

This is a standard geometric move: a metric upstairs induces a metric downstairs by pullback. In the infrared, large-scale, coarse-grained regime, ToE proposes that the corresponding curvature action reduces to an Einstein–Hilbert form plus controlled corrections:

$$I_{\text{geom}} \sim \frac{1}{16\pi G_{\text{eff}}} \int_M \sqrt{-g} (R - 2\Lambda_{\text{ent}} + \Delta_{\text{corr}}) d^4x.$$

The idea that Einstein-like dynamics can emerge from Fisher/Hessian information geometry already has nontrivial antecedents in the work of Matsueda and others; ToE's distinctive step is to insist on the Lorentzian conversion by the entropy-gradient sign-flip instead of stopping at a purely Riemannian or Euclidean emergent geometry. ¹²

A special note is needed on the **Obidi Curvature Invariant** mentioned in *Letter III*. As written there, it is introduced as

$$\text{OCI} = \ln 2,$$

interpreted as a universal lower bound on the entropic cost of distinguishing two physical states. Conceptually, this functions as a proposed **minimal distinguishability threshold**. Strictly speaking, however, it is not a curvature scalar in the ordinary differential-geometric sense, because a standard curvature invariant is usually a scalar built from the Riemann tensor and the metric, such as R , $R_{\mu\nu}R^{\mu\nu}$, or $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$. So the cleanest interpretation is this: in the manuscript, OCI is best read as a **dimensionless entropic threshold postulate** rather than as a polynomial curvature invariant. That clarification actually strengthens, rather than weakens, the theory, because it tells the reader exactly what mathematical job OCI is meant to do. [filecite turn0file0](#)

The entropic source side and the Einstein right-hand side

The user's deepest concern has been exactly right: even if information geometry can explain a metric and thus the left-hand side of Einstein's equations, how does one get the right-hand side, the stress-energy tensor? The answer is that one does **not** try to identify a single raw probability density directly with mass or pressure. Instead, one uses the same machinery that relativistic kinetic theory uses: a distribution on a bundle, integrated over fibers, produces macroscopic tensor fields. This is the cleanest mathematical road from "information distribution" to "tangible matter variables." ⁴

Let M be the emergent spacetime manifold. Above each point $x \in M$, attach a fiber F_x of microscopic entropic states—these may encode momentum-like variables, internal labels, polarization data, occupation numbers, distinguishability coordinates, or other microstate descriptors. The total space

$$E = \bigsqcup_{x \in M} F_x$$

is a **fiber bundle**, with projection $\pi : E \rightarrow M$. A field f on E is then a distribution of microscopic entropic content. The essential idea of a fiber integral is simple: integrate over the internal states at fixed spacetime point x , and you obtain an ordinary macroscopic quantity on M . That is how hidden microstructure becomes observable coarse-grained physics. ⁴

Now suppose that each microscopic state carries a momentum-like covector p_μ and that $d\mu_x$ is the invariant measure on the fiber F_x . Then the **zeroth moment** is

$$n(x) = \int_{F_x} f(x, p, \lambda) d\mu_x,$$

the **first moment** is

$$J^\mu(x) = \int_{F_x} p^\mu f(x, p, \lambda) d\mu_x,$$

and the **second moment** is

$$\Pi^{\mu\nu}(x) = \int_{F_x} p^\mu p^\nu f(x, p, \lambda) d\mu_x.$$

These moments have immediate physical meaning. The zeroth moment counts amount or density. The first moment gives net flow. The second moment gives spread, momentum flux, pressure, and stress. This is not peculiar to ToE; it is exactly how relativistic kinetic theory constructs particle current and stress-energy from a distribution function on a mass shell bundle. ⁴

That observation dissolves the central worry. A raw distribution does **not** yet look like mass or pressure; but its moments do. If the distribution is sharply localized and anchored in a rest frame u^μ , the second moment decomposes into the familiar fluid form

$$T_{\mu\nu} = \rho u_\mu u_\nu + p h_{\mu\nu} + 2u_{(\mu} q_{\nu)} + \pi_{\mu\nu},$$

where $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$ projects orthogonally to u^μ , q_μ is heat or energy flux, and $\pi_{\mu\nu}$ is anisotropic stress. In local equilibrium, $q_\mu = 0$ and $\pi_{\mu\nu} = 0$, giving the perfect-fluid tensor

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + p g_{\mu\nu}.$$

Dust corresponds to $p = 0$. Radiation corresponds to $p = \rho/3$. So mass density, pressure, radiation, momentum flux, and stress are precisely what different low-order moments of a localized microscopic distribution look like after coarse-graining. This is the clean mathematical content behind the ToE statement that “matter is concentrated entropy” or “stress-energy is organized information.” ⁴

In ToE, therefore, the Einstein right-hand side should be written not as a single monolithic object but as a sum of entropic sectors. The most natural decomposition is

$$\mathbb{T}_{\mu\nu}^{\text{ToE}} = T_{\mu\nu}^{(S)} + T_{\mu\nu}^{(\text{fib})} + T_{\mu\nu}^{(\text{constr})}.$$

Here $T_{\mu\nu}^{(S)}$ is the stress tensor of the entropy field itself, $T_{\mu\nu}^{(\text{fib})}$ is the coarse-grained fiber-moment tensor of localized entropic microstructure, and $T_{\mu\nu}^{(\text{constr})}$ collects auxiliary or constraint sectors such as the G-field discussed in *Letter III*. A canonical scalar-field contribution would have the familiar form

$$T_{\mu\nu}^{(S)} = \chi \left(\nabla_\mu S \nabla_\nu S - \frac{1}{2} g_{\mu\nu} (\nabla S)^2 \right) - g_{\mu\nu} V(S),$$

while

$$T_{\mu\nu}^{(\text{fib})} = \int_{F_x} p_\mu p_\nu f d\mu_x$$

is the direct entropic analogue of the kinetic-theory stress-energy tensor. This is the mathematically appropriate “entropic rank-two tensor” that stands underneath the ordinary Einstein source tensor in the ToE picture. [filecite\[turn0file0\]](#)

The full effective gravitational equation then takes the natural form

$$G_{\mu\nu}[g] + \Lambda_{\text{ent}} g_{\mu\nu} = 8\pi G_{\text{eff}} \mathbb{T}_{\mu\nu}^{\text{ToE}}.$$

This formula says exactly what the user has been trying to express conceptually: the standard stress-energy tensor is the low-energy, coarse-grained limiting case of a more primitive entropic source structure. In that limit,

$$G_{\text{eff}} \rightarrow G, \quad \Lambda_{\text{ent}} \rightarrow \Lambda, \quad \mathbb{T}_{\mu\nu}^{\text{ToE}} \rightarrow T_{\mu\nu}^{\text{Einstein}},$$

so ordinary general relativity is recovered. The key conceptual gain is that the right-hand side is no longer “put in by hand” as a separate matter sector alien to information; it is produced by the fiber-integrated statistics of localized entropic microstructure. [filecite\[turn0file0\]](#)

Conservation is the final consistency test. On the geometric side, the Bianchi identity gives

$$\nabla^\mu G_{\mu\nu} = 0.$$

Therefore any acceptable source must satisfy

$$\nabla^\mu \mathbb{T}_{\mu\nu}^{\text{ToE}} = 0.$$

In relativistic kinetic theory this happens when the distribution function satisfies a Liouville or Boltzmann equation whose microscopic interactions conserve momentum: fiber integrals then produce divergence-free current and stress-energy tensors. The same logic can be imported into ToE. If the entropic distribution evolves by a covariant kinetic equation preserving the relevant micro-conservation laws, then the resulting entropic source tensor will automatically satisfy the local accounting identity required by the Einstein side. This is the rigorous mathematical core of what *Letter III* calls entropic accounting. ¹³

What ToE has achieved mathematically and what remains open

The strongest part of ToE, mathematically, is not the slogan that “everything is entropy.” The strongest part is the realization that the route from entropy to physics must pass through three precise structures: **divergence geometry, signature-changing metric deformation, and fiber-moment coarse-graining.** Once those are in place, the theory stops sounding mystical and starts sounding like a sharp research program. The Fisher–Rao and Fubini–Study metrics supply distinguishability geometry; the Obidi transformation provides a candidate mechanism for converting that geometry into Lorentzian causal structure; and fiber integrals provide the correct mathematical method for turning localized entropic distributions into rank-two source tensors. That combination gives a clear conceptual answer to the user’s central question: information becomes “tangible” not by magic, but by **organization, localization, and moment extraction.** ¹⁴

It is also now possible to say exactly where ToE is original. Jacobson showed how Einstein dynamics can emerge thermodynamically from horizon entropy. Verlinde argued for gravity as entropic force. Matsueda derived Einstein-like structures from Fisher information. Bianconi proposed a gravity action built from quantum relative entropy. ToE’s distinctive move is to insist that **both** the geometric left-hand side and the entropic-matter right-hand side should arise from a single informational substrate, and that the crucial missing bridge from ordinary information geometry to relativity is the emergence of Lorentzian signature through an entropy-selected direction. That is the conceptual niche ToE occupies. ¹⁴

At the same time, several parts of the program remain open and should be stated honestly. The exact microscopic identity of the fiber variables F_x is not yet fixed. The fundamental action from which the Obidi metric, the entropy field, and the source sectors all descend in a unique way still needs a fully explicit and universally agreed final form. The global existence conditions for the Obidi transformation—especially at points where $dS = 0$ —need a careful treatment. The Status of the OCI should be clarified terminologically, as explained above. And the theory still needs a mature phenomenology: concrete predictions distinguishing ToE from general relativity, scalar-tensor gravity, entropic gravity, and quantum-information-inspired gravity programs. These are not fatal weaknesses; they are the natural open problems of a young foundational program. ¹⁵

The proper conclusion is therefore this. ToE is **most compelling** when it is formulated not as a vague declaration that “entropy explains everything,” but as a disciplined chain of mathematical claims:

entropy/divergence $\implies$ information geometry $\implies$ Lorentzian metric by entropy-gradient deformation $\implies$ causal structure
microscopic entropic distribution on fibers $\implies$ zeroth/first/second moments $\implies$ entropic rank-two source tensors

When stated that way, the theory becomes understandable even to a beginner: the left-hand side comes from how distinguishability is turned into geometry, and the right-hand side comes from how localized entropic microstructure is turned into matter-like moments. That is the mathematical heart of the Theory of Entropicity. [filecite turn0file0](#)

Open questions and limitations

The clearest open questions are four. First, what is the exact microscopic configuration space whose fibers are being integrated: classical phase-space variables, quantum state labels, or a unified entropic microstate bundle? Second, can the Obidi transformation be derived from a deeper variational principle rather than postulated? Third, can G , c , and $\hbar$ be recovered as emergent couplings from the same entropic action? Fourth, can the theory produce observational signatures beyond standard GR, such as modified lensing, cosmological corrections, decoherence thresholds, or thermodynamic bounds tied to the OCI? These questions are precisely where the next technical development of ToE should focus. [filecite turn0file0](#)

References

The principal ToE source for this introduction is your attached *Theory of Entropicity Letter III*, which states the program in its native form, introduces the Obidi transformation and metric, proposes the α - q constitutive relation, introduces the G-field and dressed Einstein equations, and defines the OCI threshold.

[filecite turn0file0](#)

Foundational external references used here include: Ted Jacobson, *Thermodynamics of Spacetime: The Einstein Equation of State*; Erik Verlinde, *On the Origin of Gravity and the Laws of Newton*; Ariel Caticha and collaborators on entropic dynamics on statistical manifolds; Hiroaki Matsueda on emergent general relativity from Fisher information; Ginestra Bianconi, *Gravity from Entropy* and its related quantum-relative-entropy developments; Olivier Sarbach and Thomas Zannias on relativistic kinetic theory, tangent bundles, and fiber-integral constructions of current and stress-energy; and standard information-geometric and quantum-informational definitions of Fisher–Rao, Fubini–Study, Kullback–Leibler, and Umegaki relative entropy. ¹⁵

¹ ² ¹⁴ ¹⁵ <https://arxiv.org/abs/gr-qc/9504004>

<https://arxiv.org/abs/gr-qc/9504004>

³ <https://arxiv.org/abs/1611.07546>

<https://arxiv.org/abs/1611.07546>

⁴ ⁹ ¹³ <https://arxiv.org/abs/1303.2899>

<https://arxiv.org/abs/1303.2899>

⁵ https://en.wikipedia.org/wiki/A_Mathematical_Theory_of_Communication

https://en.wikipedia.org/wiki/A_Mathematical_Theory_of_Communication

⁶ <https://arxiv.org/abs/1705.01064>

<https://arxiv.org/abs/1705.01064>

⁷ https://en.wikipedia.org/wiki/Fubini%E2%80%93Study_metric

https://en.wikipedia.org/wiki/Fubini%E2%80%93Study_metric

8 https://en.wikipedia.org/wiki/Quantum_relative_entropy
https://en.wikipedia.org/wiki/Quantum_relative_entropy

10 <https://arxiv.org/abs/gr-qc/9211017>
<https://arxiv.org/abs/gr-qc/9211017>

11 https://en.wikipedia.org/wiki/Contracted_Bianchi_identities
https://en.wikipedia.org/wiki/Contracted_Bianchi_identities

12 **Emergent General Relativity from Fisher Information Metric**
https://arxiv.org/abs/1310.1831?utm_source=chatgpt.com